# Root Grip Erosion Rescue

Can roots hold a slope together during a storm?

**Win condition:** Most soil retained after standard pour.

## THE SCIENCE YOU ARE USING

**Roots anchor soil.** Bare soil on a slope washes away fast when rain hits it. Roots grip the soil and hold it in place, and a denser, deeper root network holds better. This is why planting vegetation is a real tool for stopping erosion.

**Slope and runoff.** Steeper slopes and bare surfaces shed water and soil quickly. You build a mini slope with root-like anchors, pour a standard amount of water, and measure how much soil stays put.

## HOW TO BUILD AND RUN IT

01 **Build a bare slope.** Pack soil into a tray at a set angle with no roots. This is your control.

02 **Add root anchors.** Push string or yarn roots into a second slope to mimic a planted hillside.

03 **Pour the standard storm.** Pour the same measured water on each slope from the same height.

04 **Measure soil retained.** Compare how much soil washed out of each tray. Record the difference.

05 **Redesign the root network.** Add more or deeper roots and re-test to retain more soil.

**Safety:** Work over a tray and wear gloves to handle wet soil and muddy water; wipe spills right away so the floor stays dry. Pour muddy water into the slop bucket, not the sink. No goggles are needed for pouring water down a soil tray. Allergy: Dust or pollen sensitivity, so the soil is sieved and dampened to keep dust down.

## MATERIALS

| | |
|---|---|
| Trays | 1 per team (4) |
| Soil | shared |
| String or yarn roots | shared |
| Craft sticks | shared |
| Watering cups | 1 per team (4) |

## YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

**Prediction (what will work best, and why):** ______

**The ONE thing we changed for the redesign (more roots, deeper roots, or a denser pattern):** ______

**Soil retained, baseline (bare vs rooted slope):** ______

**Soil retained, after redesign:** ______

## DEFEND YOUR DESIGN

**What evidence made you change your design?** ______

**If you ran it again, what is the ONE thing you would change?** ______

## HOW YOU SCORE (100 POINTS)

| SCORING CRITERION | POINTS |
|---|---|
| Soil retained | 40 |
| Runoff control | 25 |
| Design explanation | 20 |
| Cleanup | 15 |
| **Total** | **100** |